

16S rRNA gene compositional analysis provides a summary of the composition and structure of the bacterial component of the microbiome. Genomic bacterial DNA extraction methods were optimized to maximize the yield of bacterial DNA while keeping background amplification to a minimum. 16S rRNA gene sequencing methods were adapted from the methods developed for the Earth Microbiome Project (X) and NIH-Human Microbiome Project (1,2,3). Briefly, bacterial genomic DNA was extracted from the total mass of the biopsied specimens using the MoBio PowerLyzer Tissue and Cells DNA isolation kit and sterile spatulas for tissue transfer. The 16S rDNA V4 region was amplified from the extracted DNA by PCR and sequenced in the MiSeq platform (Illumina) using the 2x250 bp paired-end protocol yielding pair-end reads that overlap almost completely. The primers used for amplification contain adapters for MiSeq sequencing and single-index barcodes so that the PCR products may be pooled and sequenced directly(3), targeting at least 10,000 reads per sample.

CMMR 16S (variable region 4 [v4]) rRNA gene pipeline data incorporates phylogenetic and alignment based approaches to maximize data resolution. The read pairs are demultiplexed based on unique molecular barcodes added via PCR during library generation, then merged using USEARCH v7.0.1090 (4). The subsequent analysis steps of the pipeline leverage custom analytic packages developed at the Alkek Center for Metagenomics and Microbiome Research (CMMR) at Baylor College of Medicine to produce summary statistics and quality control measurements for each sequencing run, as well as multi-run reports and data-merging capabilities for validating built-in controls and characterizing microbial communities across large numbers of samples or sample groups. 16Sv4 rDNA sequences are clustered into Operational Taxonomic Units (OTUs) at a similarity cutoff value of 97% using the UPARSE algorithm (5). OTUs are subsequently mapped to an optimized version of the SILVA Database (6) containing only sequences from the v4 region of the 16S rRNA gene to determine taxonomies. Abundances are recovered by mapping the demultiplexed reads to the UPARSE OTUs. A custom script constructs an OTU table from the output files generated in the previous two steps for downstream analyses using a visualization toolkit also developed at the CMMR named ATIMA (*Agile Toolkit for Incisive Microbial Analyses*). ATIMA is a stand-alone tool for analyzing and visualizing microbiome data sets. This software provides an integrated solution for exploring relationships between microbial communities and emergent properties of their hosts or environments. Through this web portal, data (biom files and supporting files) are uploaded to generate interactive plots visualizing trends in alpha diversity (richness and evenness), beta diversity (in-between sample differences), and taxa abundances (phylum-genus) through customizable box plots, Principal Coordinate Analysis (PCoA) ordinations, and heatmaps. Statistical annotations are added to denote significant correlations with metadata, enabling quick assessment of many variables. Generated data and figures can be exported as Excel-compatible spreadsheets and publication quality PDFs. These features promote in-depth interrogation of microbial communities by researchers across a wide range of expertise.

ATIMA is an R (8) software suite combining publicly available packages (i.e. APE and VEGAN) (9,10) and purpose written code to import sample data and identify trends in taxa abundance,

alpha-diversity, and beta-diversity with sample metadata. Significance of categorical variables are determined using the non-parametric Mann-Whitney test (11) for two category comparisons or the Kruskal-Wallis test (12) when comparing three or more categories. Correlation between two continuous variables is determined with R's base "lm" function for linear regression models, where p-values indicate the probability that the slope of the regression line is zero. PCoA plots employ the Monte Carlo permutation test (13) to estimate p-values. All p-values are adjusted for multiple comparisons with the FDR algorithm (14).

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